# Cantor's Theorem and the Power-Set Hierarchy

## An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

For any proposed enumeration of subsets, choose a new subset that disagrees with the assigned subset at each element's own membership test. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**LF-01 / Mathematical Logic and Set Theory / Report v1**

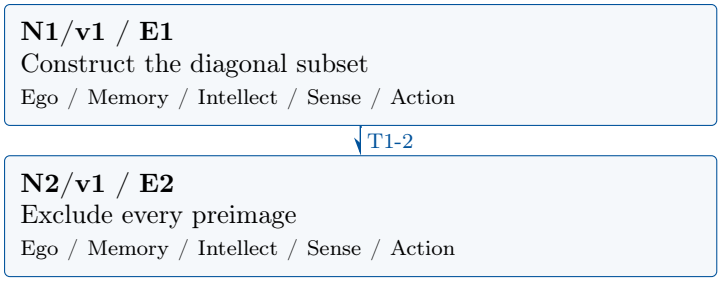
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *For every set $A$, prove there is no surjection $f : A \to \mathcal{P}(A)$, where $\mathcal{P}(A)$ is the set of all subsets of $A$. Since $a \mapsto \{a\}$ is an injection, conclude that $A$ has strictly smaller cardinality than its power set, in the sense of injection plus absence of a bijection. The diagonal argument does not require the axiom of choice.*

### A. Foundations and permitted imports

Use the power-set and separation axioms of ordinary set theory, the definitions of injective, surjective, and bijective functions, and classical propositional logic. Separation permits forming a subset of an already given set by a formula. No unrestricted "set of all sets" comprehension is used.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1-2: N1/E1/v1 to N2.** $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

**Receiving activity.** N2.1

**Compatibility.** E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Check: The contradiction applies to every proposed surjection, including finite and empty domains.

## III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

### A. N1/v1: Construct the diagonal subset

**Object and role.** Use bounded separation to reverse each element's own membership decision.
**Local obligation:** $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

**Inputs.** An arbitrary function $f : A \to \mathcal{P}(A)$.

**E — Ego**

Publish E1/v1 only after establishing this obligation: $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

**M — Memory**

An arbitrary function $f : A \to \mathcal{P}(A)$. Retain the stated assumptions and import versions.

**I — Intellect**

Use bounded separation to reverse each element's own membership decision.

**S — Sense**

The definition produces a subset of $A$, so it belongs to the codomain; no self-containing universal set is needed.

**A — Action**

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N1.1 For each $a \in A$, $f(a)$ is a subset of $A$, so the membership statement $a \in f(a)$ is meaningful. By separation, the elements of $A$ for which this statement is false form a set $D \subset A$.

N1.2 The definition gives, for each $a \in A$, the exact equivalence $a \in D \iff a \notin f(a)$. In particular $D$ is in the stated codomain $\mathcal{P}(A)$.

**Export E1/v1.** $D = \{a \in A : a \notin f(a)\}$ is a well-defined member of $\mathcal{P}(A)$.

**Acceptance gate.** The definition produces a subset of $A$, so it belongs to the codomain; no self-containing universal set is needed.

**Targeted repair.** If the construction is made over all sets rather than $A$, restrict it to $A$; bounded separation is the actual permitted axiom. Withhold this export and recheck its descendants after a versioned correction.

## B. N2/v1: Exclude every preimage

**Object and role.** Specialize the membership equivalence at the alleged preimage and derive a contradiction. **Local obligation:** No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

**Inputs.** E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Incoming proof exports: E1/v1 at N2.1.

**E — Ego**

Publish E2/v1 only after establishing this obligation: No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

**M — Memory**

E1/v1 and a hypothetical $b \in A$ with $f(b) = D$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

**I — Intellect**

Specialize the membership equivalence at the alleged preimage and derive a contradiction.

**S — Sense**

The contradiction applies to every proposed surjection, including finite and empty domains.

**A — Action**

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N2.1 If $f(b) = D$, the equivalence in E1 reads $b \in D \iff b \notin D$, impossible in classical logic. Thus no element of $A$ maps to $D$, and the arbitrary function $f$ is not surjective.

N2.2 The singleton map $a \mapsto \{a\}$ is injective because equal singleton sets have equal elements. A bijection would in particular be a surjection, already ruled out. This proves the claimed strict cardinal comparison. For $A = \varnothing$, $\mathcal{P}(A) = \{\varnothing\}$ and the same lack of surjection is immediate.

**Export E2/v1.** No $f : A \to \mathcal{P}(A)$ is surjective; the singleton map establishes strict cardinal growth.

**Acceptance gate.** The contradiction applies to every proposed surjection, including finite and empty domains.

**Targeted repair.** If a contradiction is interpreted as showing the power set is inconsistent, identify the rejected assumption: only the proposed surjectivity fails. Withhold this export and recheck its descendants after a versioned correction.

## IV. CONTINUOUS MATHEMATICAL PROOF

### A. N1 — Construct the diagonal subset

1. **N1.1.** For each $a \in A$, $f(a)$ is a subset of $A$, so the membership statement $a \in f(a)$ is meaningful. By separation, the elements of $A$ for which this statement is false form a set $D \subset A$.

2. **N1.2.** The definition gives, for each $a \in A$, the exact equivalence $a \in D \iff a \notin f(a)$. In particular $D$ is in the stated codomain $\mathcal{P}(A)$.

### B. N2 — Exclude every preimage

1. **N2.1.** If $f(b) = D$, the equivalence in E1 reads $b \in D \iff b \notin D$, impossible in classical logic. Thus no element of $A$ maps to $D$, and the arbitrary function $f$ is not surjective.

2. **N2.2.** The singleton map $a \mapsto \{a\}$ is injective because equal singleton sets have equal elements. A bijection would in particular be a surjection, already ruled out. This proves the claimed strict cardinal comparison. For $A = \varnothing$, $\mathcal{P}(A) = \{\varnothing\}$ and the same lack of surjection is immediate.

□

## V. WORKED ORCHESTRATION

For $A = \{1, 2, 3\}$, consider $f(1) = \{1, 2\}$, $f(2) = \{1\}$, and $f(3) = \{3\}$. The diagonal rule includes 2 but excludes 1 and 3, so $D = \{2\}$. It is different from all three listed images, illustrating the exact self-index disagreement.

### A. Boundary and hypothesis audit

Listing all finite subsets of $\mathbb{N}$ is possible and does not contradict Cantor: the codomain is then only a proper part of $\mathcal{P}(\mathbb{N})$. The diagonal subset for such a list cannot be one of its listed finite subsets. Replacing "all subsets" by "finite subsets" changes the target.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Rejected argument, diagnosis, and correction

**Rejected scope (N1):** form the set of all sets that do not contain themselves. **Diagnosis:** Memory replaced separation over $A$ by unrestricted comprehension. **Repair:** construct only $D \subset A$ using the given map; its existence is licensed by separation. Then N2 contradicts surjectivity, not the set axioms. Recheck the codomain membership and final cardinal comparison.

### B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Apply the construction to $A = \mathbb{N}$ to obtain an uncountable power set and to explain why no single list can enumerate all binary sequences.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.** Andrew Marks, Set Theory notes for sets and ultraproducts; Berkeley, Mathematical Logic for Mathematicians, Part I for first-order semantics; Andrew Marks, Computability Theory notes for effective procedures and undecidability. The proofs below specify their own set-theoretic and semantic assumptions.